

PROJECT PROPOSAL

PHASE 1



E-RANK: DIGITAL TAXI RANK MANAGEMENT AND PASSENGER SAFETY SYSTEM

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Introduction

The minibus taxi industry is the dominant form of public transport in South Africa and plays a critical role in the daily mobility of millions of people. Studies indicate that a large percentage of commuters rely on minibus taxis to travel to work, school, and other destinations (Africa Check, 2021). Taxi ranks are therefore important environments where transport coordination takes place.

At taxi ranks, the taxi marshal is responsible for maintaining order and controlling daily operations. The marshal directs passengers to taxis, manages taxi queues, collects fares, records passenger details for long-distance travel, and informs drivers when taxis are full and ready to depart. In many taxi ranks, this process is performed manually using notebooks, verbal communication, and informal queue systems.

This manual system creates several challenges. Passenger information recorded for long-distance travel can be lost or damaged because it is stored in physical logbooks. Taxi queue management can also lead to disputes between drivers because the order of taxis is not digitally recorded. Taxi owners have limited visibility into the number of trips completed by their taxis each day, making it difficult to monitor operations or revenue.

Passengers also experience uncertainty because information about taxi availability, routes, and operating hours is not always clearly communicated. These issues affect the overall user experience for passengers, drivers, taxi owners, and marshals.



Problem Statement

The current taxi rank management system relies heavily on manual processes handled by taxi marshals. Passenger details for long-distance trips are often written in physical record books, which can easily be lost or damaged. Taxi queue management is also done manually, which can lead to disputes between drivers regarding queue positions. In addition, taxi owners have limited visibility of how many trips their vehicles complete each day.

Passengers often experience uncertainty regarding taxi availability, routes, and operating hours. Communication between drivers, passengers, and taxi owners is limited, which results in long waiting times and inefficient taxi departures. These challenges highlight the need for a digital system that can support taxi rank operations while maintaining the existing workflow managed by the taxi marshal.

Several initiatives have attempted to introduce digital technology into the South African taxi industry. Examples include the **Loop Taxi platform**, the **WOW (Wealth on Wheels) digital taxi initiative**, and the **FairPay transport payment system**.

The **Loop Taxi platform** was developed to help commuters book taxi rides through a mobile application and allow drivers to receive digital payments. Although the system aimed to improve

passenger access to taxis and increase driver income during off-peak periods, it did not achieve widespread adoption because it required passengers and drivers to rely heavily on smartphone technology. Many taxi operations still depend on physical taxi ranks and marshal coordination rather than digital ride booking.

The **WOW (Wealth on Wheels) initiative** introduced technology such as GPS tracking, onboard cameras, Wi-Fi, and fleet monitoring tools for taxis. While this system improved safety and vehicle monitoring, it focused mainly on technology inside the taxi vehicle rather than on **taxi rank operations**, where passenger coordination and queue management take place.

Similarly, the **FairPay transport card system** aimed to introduce cashless payment methods for public transport. While this system improved payment convenience, it did not address operational challenges such as taxi queue disputes, passenger registration for long-distance travel, and communication between marshals and drivers.

These systems demonstrate that digital innovation has been introduced in the taxi industry, but many of them failed to fully address the **rank-level coordination managed by taxi marshals**. Because taxi ranks remain the central point of passenger movement and taxi organization, a system that supports taxi marshal operations is needed.



Proposed Solution

To address these challenges, this project proposes a digital taxi rank management system called **E-RANK**.

The system is designed to support the **taxi marshal as the primary controller of taxi rank operations**. Through a tablet or mobile device, the marshal will be able to manage taxi queues, register passengers for long-distance trips, record fares collected from passengers, and notify drivers when taxis are ready to depart.

Drivers will interact with the system by scanning a QR code to register their position in the taxi queue and confirming trips once they are completed. Taxi owners will be able to monitor their taxis through a dashboard that displays how many trips each taxi has completed. Passengers will be able to receive information about routes, taxi availability, and travel updates.

The key gap between the current system and the proposed system is that the current taxi rank operations rely on manual processes, while the proposed E-RANK system introduces digital tools for **queue management, passenger registration, trip tracking, and communication**.

The E-RANK system improves the user experience for all stakeholders within the taxi rank environment.

For taxi marshals, the system simplifies operational tasks by providing digital tools for queue management, passenger registration, and fare recording. This reduces administrative workload and improves efficiency.

For drivers, the system reduces queue disputes by providing a clear digital queue system and allows drivers to confirm completed trips easily.

For taxi owners, the system provides better visibility into taxi activity by allowing them to monitor the number of trips completed by each vehicle.

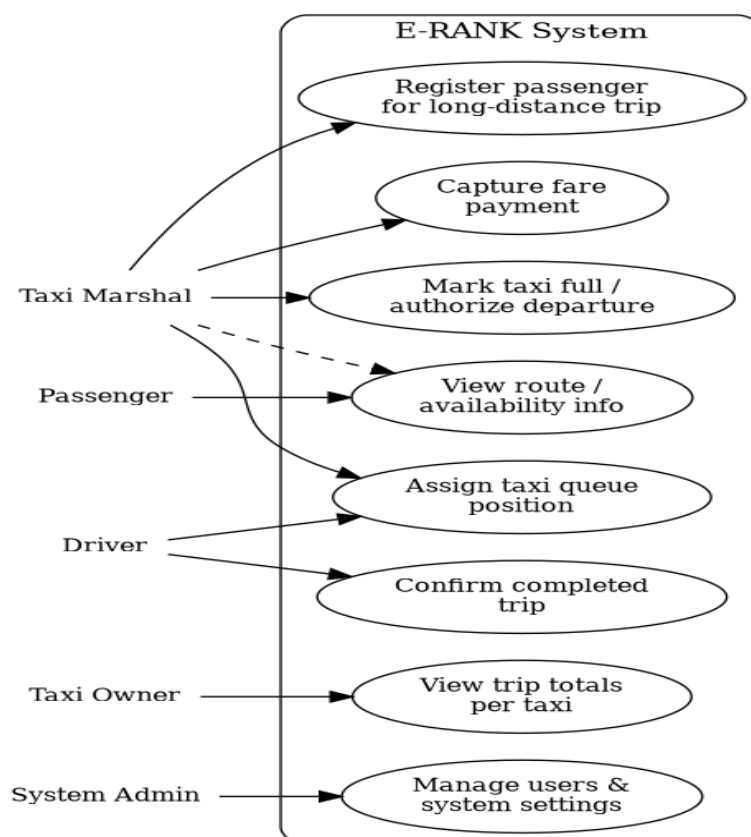
For passengers, the system improves communication and safety by providing better information about taxi routes, availability, and travel records.

Overall, the E-RANK system enhances **efficiency, transparency, safety, and communication** within the taxi rank ecosystem.



System Actors

The E-RANK system involves several key actors who interact with the platform. The taxi-marshall is the primary actor responsible for managing taxi rank operations. Passengers interact with the system to receive information about routes and taxi availability. Drivers use the system to join the taxi queue and confirm completed trips. Taxi owners use the system to monitor taxi activity and performance.





System Functionalities

- The system will allow drivers to scan a QR code to join the taxi queue.
- The system will allow marshals to manage the taxi queue digitally.
- The system will allow marshals to register passenger details for long-distance trips.
- The system will store passenger emergency contact information.
- The system will notify drivers when a taxi is full and ready to depart.
- The system will allow marshals to record fare collection.
- The system will allow drivers to confirm completed trips.
- The system will allow taxi owners to monitor the number of trips completed by their vehicles.
- The system will provide route and destination information to passengers.
- The system will display taxi availability information.
- The system will send SMS alerts regarding taxi departures.
- The system will provide real-time updates about taxi rank activity.

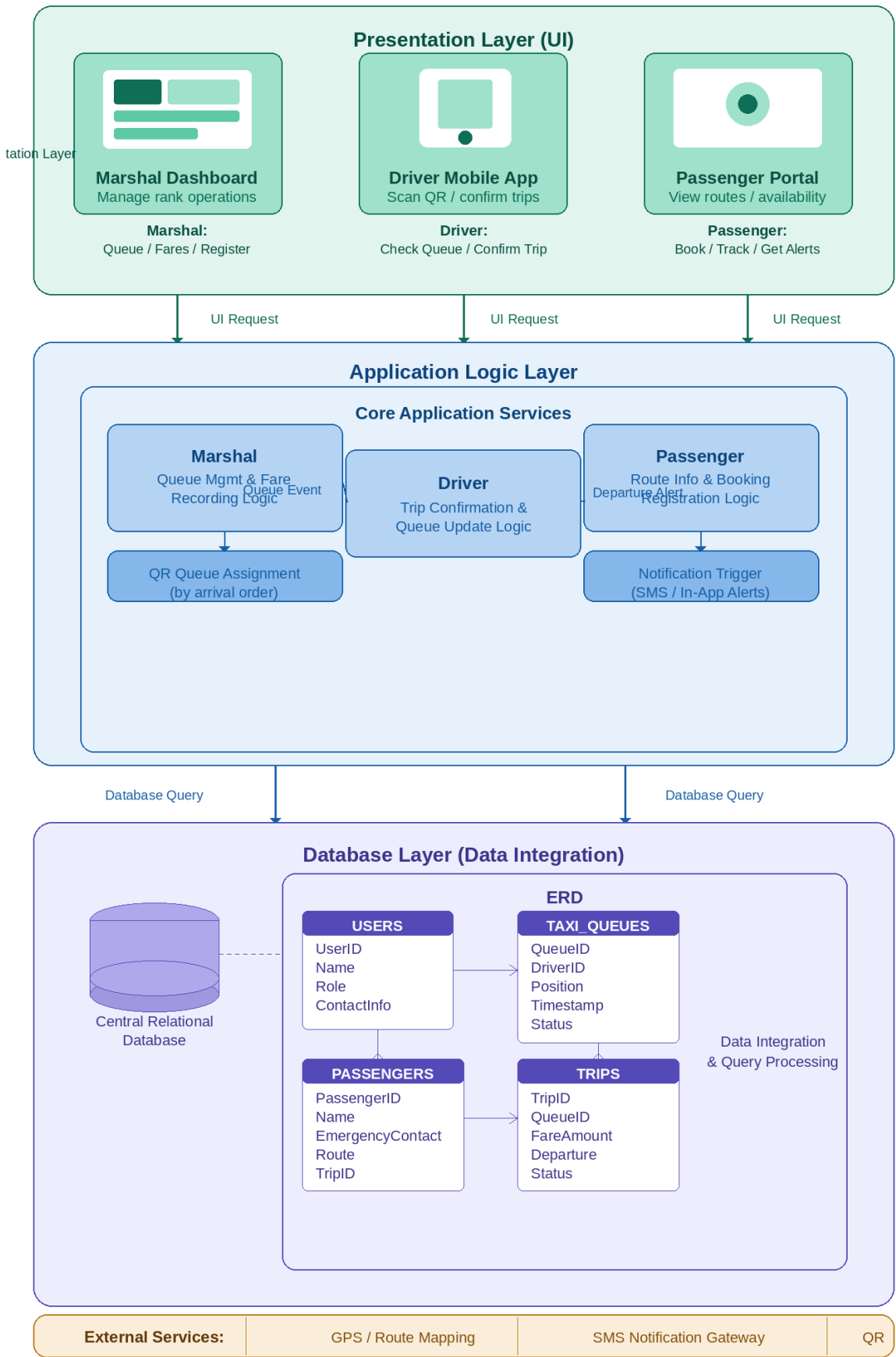


System Architecture

The E-RANK system follows a layered architecture design. The user layer consists of passengers, drivers, and taxi marshals who interact with the system using mobile devices or tablets. The user interface layer provides screens for passenger information, marshal controls, and driver queue registration. The application logic layer processes core system operations such as queue management, passenger registration, and trip management. The data layer stores information about passengers, taxi queues, and trips. External services such as GPS mapping and SMS notifications support location tracking and communication.

E-RANK: Digital Taxi Rank Management System

System Architecture





Conclusion

The proposed E-RANK system provides a practical digital solution for improving taxi rank operations. By focusing on the role of the taxi marshal as the central controller of rank activities, the system enhances efficiency, transparency, and passenger safety. The platform also allows taxi owners to monitor their vehicles and helps drivers manage their trips more effectively.



REFERENCES

Africa Check. 2021. *Do more than 80% of South Africans rely on minibus taxis?* Available at: <https://africacheck.org> (Accessed: 1 March 2026).

Arrive Alive. 2024. *Minibus taxis and road safety in South Africa*. Available at: <https://www.arrivealive.mobi> (Accessed: 1 March 2026).

Department of Transport. 2024. *Annual Transport Statistics Report*. Pretoria: Department of Transport, Republic of South Africa.

International Transport Forum. 2023. *Digitalisation of transport systems and mobility services*. Paris: Organisation for Economic Co-operation and Development (OECD).

Santaco. 2023. *Transformation and digital innovation in the minibus taxi industry*. Available at: <https://www.santaco.co.za> (Accessed: 24 March 2026).

Kisten, M. 2026. *NPRT63 ICT Project Lecture Notes*. Sol Plaatje University.

BBC News. 2026. *Violence and safety challenges facing e-hailing drivers in South Africa*. Available at: <https://www.bbc.com> (Accessed: 1 March 2026).

Daily Maverick. 2022. *New app for minibus taxis aims to improve transport services*. Available at: <https://www.dailymaverick.co.za> (Accessed: 1 March 2026).